

Steering a Network with Its Own Reflection: True, False, and Noise Self-Model Broadcasts in an Evolving Multi-Agent Network

Ivan Bhargava

Independent Researcher

`ivan.bhargava@example.com`

WWW home page: <https://github.com/Ivan825/Project-ONE>

Abstract. Many networks respond to measurements of their own state: markets to indices, organizations to metrics, platforms to trending signals. We present a controlled laboratory for this reflexive loop: an evolving agent population forms an adaptive network whose macrostate is compressed into a self-model and broadcast back — accurately, distorted, or as matched-bandwidth noise. Across 927 deterministic runs with paired seeds and pre-specified outcomes: measurement without broadcast is causally inert; false self-models pull the macrostate toward themselves far more strongly than matched noise (paired rank-biserial $r = 0.84\text{--}0.88$), and steering power grows with the plausibility of the description — a replayed genuine self-model from another run steers hardest of all; which lies come true depends on the response regime, an alarmist lie flipping from self-defeating under corrective responses to strongly self-fulfilling under conformist ones; and heritable attention to the broadcast is selected against in proportion to its unreliability: populations fed false self-models evolve to ignore their observer.

Keywords: adaptive networks, agent-based modeling, reflexivity, self-fulfilling dynamics, evolution of trust, network resilience

1 Introduction

Adaptive networks — graphs whose topology coevolves with the dynamical states of their nodes — are a central object of network science [1, 2]. A distinctive subclass has received far less controlled attention: networks that respond to *descriptions of themselves*. Financial markets move on published indices computed from their own trades [3]; universities restructure around rankings derived from their own behavior [4]; recommendation platforms amplify items *because* an internal measurement declared them trending. In each case a macroscopic measurement of the system is compressed, published, and re-enters the system as an input to microscopic decisions — a loop Merton called the self-fulfilling prophecy [5] and whose measure-distorting variant is known as Goodhart’s law [6].

Empirical study of this loop is confounded: in real systems one can rarely withhold the measurement, falsify it in controlled directions, or replay history

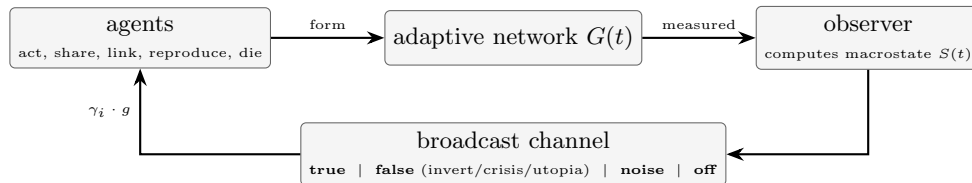


Fig. 1. The reflexive loop and its experimental corruption points. Agents’ local actions generate the network; the observer compresses it into a self-model; the broadcast channel returns that self-model truthfully, distorted, as noise, or not at all. Each agent’s response is scaled by its heritable sensitivity γ_i and a global gain g .

under identical randomness. We therefore built a minimal simulated world in which the loop can be opened, closed, and corrupted at will, and where every run is exactly reproducible from a configuration and seed. Agents are deliberately simple — transparent trait vectors rather than learned policies — so that every behavioral pathway from broadcast to response is inspectable.

Our contributions are: (1) an open, deterministic experimental platform coupling an evolving agent ecology, an adaptive interaction network, a global observer, and a controllable broadcast channel; (2) a five-condition causal design that separates the effect of *being measured* from the effect of *being told*, and the effect of *self-information* from that of *stimulation* (via a matched-bandwidth noise control); (3) evidence, robust across feedback gains and two shock severities, that false self-models steer the macrostate toward themselves while noise does not, and that the steering direction is a function of the distortion type; and (4) to our knowledge the first demonstration in an ecological agent-based setting that *attention to a collective self-model is itself an evolvable trait*, selected against in proportion to the self-model’s unreliability. Figure 1 summarizes the design.

2 Related Work

Adaptive and temporal networks. Coevolution of state and topology is surveyed in [1, 2]; our model belongs to this family, with births, deaths and rewiring driven by agent decisions. Robustness of networks to random and targeted node removal is classical [7]; we use targeted hub removal as a standardized perturbation rather than an object of study. **Global coupling in physics.** Globally coupled oscillators and maps show that a mean field fed back to units reshapes collective dynamics [8]. Our design differs in two ways central to the research question: the “mean field” is a multidimensional, potentially *false* description, and units are born, die, and evolve. **Reflexivity and reactive measurement.** Self-fulfilling prophecy [5], reflexivity in markets [3], and the reactivity of rankings and metrics [4, 6] motivate our false-broadcast conditions; we contribute a platform where these effects are dosable and ablatably. **Signaling and the evolution of trust.** In sender–receiver games, receivers can evolve to ignore un-

reliable signals [9]. Our distrust result is consonant with that theory but arises in an open ecological population where the signal is *self-referential* — a compressed description of the receiving population itself — and where attention to it is a continuous heritable trait under selection from an energy economy, not a payoff matrix. **Agent-based modeling.** Agent-based studies of collective opinion and cooperation are a mainstay of this community [16, 17]; our methodology follows the generative ABM tradition [10, 11]; the engine uses ideas from Mesa [12] and NetworkX [13]. Multi-agent reinforcement learning studies global state as an *architectural* aid [14]; we instead treat global information as an *experimental variable* with controlled veracity. Populations of language-model agents are now deployed with shared dashboards describing their own collective state [15]; the present rule-based setting provides causally clean baselines for that emerging regime.

3 The Model

3.1 Agents and ecology

Each agent i carries a heritable trait vector $\theta_i = (\text{coop}, \text{expl}, \text{risk}, \text{soc}, \text{share}, \gamma_i) \in [0, 1]^6$, where γ_i is its *global sensitivity* — how strongly broadcasts modulate its behavior. Agents hold energy; each step they pay metabolism and choose one action — harvest, share (costly transfer to a neighbor), connect, prune, reproduce, or idle — sampled with probabilities proportional to trait- and state-dependent propensities. The resource pool regenerates logistically, giving density-dependent competition. Reproduction requires an energy threshold and copies traits with Gaussian mutation ($\sigma = 0.05$); energy exhaustion or age causes death. Founding ages are staggered to avoid cohort artifacts.

3.2 Observer and broadcast conditions

Every $\Delta t = 10$ steps the observer computes the macrostate $S(t)$: population, fragmentation (1 minus largest-component share), degree- and betweenness-concentration, Freeman centralization [19], per-capita costly-helping rate, energy Gini, and turnover, plus mean trait values (logged, never broadcast). Five conditions define what agents receive (Table 1). Distorted broadcasts come in three modes: *invert* ($b_k = 1 - s_k$ per normalized metric), *crisis* (fixed alarming values), and *utopia* (fixed reassuring values). Broadcast values shift action propensities through fixed corrective response rules — e.g., reported fragmentation above 0.3 raises the propensity to form links; a reported cooperation deficit raises the propensity to share — each scaled by γ_i and a global gain g .

3.3 Emergent network structure

The interaction networks that emerge are non-trivial: at $t = 2,000$ the giant component exhibits clustering $1.8\text{--}2.4\times$ an Erdős–Rényi null with matched size

Table 1. Experimental conditions. All start from matched parameter distributions with paired random seeds.

Condition	Observer	Agents receive	Role
A — Local only	off	nothing	baseline
B — Observed, blind	on	nothing	controls for measurement
C — True feedback	on	accurate $S(t)$	main treatment
F — False feedback	on	distorted $S(t)$	reflexivity probe
N — Noise feedback	on	uniform random, same keys/ranges	stimulation control
R — Replayed self-model	on	genuine $S(t)$ trajectory of another run	structured-signal control

and density, at comparable characteristic path lengths ($L \approx 1.9$ – 2.5) — a small-world-like regime — with a broad degree spread (coefficient of variation 0.6–0.75; maximum degree 2– $2.5 \times$ the mean). The substrate on which feedback acts is therefore a heterogeneous, clustered network rather than a homogeneous mixing population.

3.4 Reproducibility

The simulation is deterministic given (configuration, seed): one PRNG drives all choices, agents act in sorted order, and a state hash certifies replay. Identical hashes were verified across operating systems and machines. All code, configurations, and analysis scripts are public; every figure regenerates from two commands.

4 Experimental Design

Four outcomes were fixed before the first campaign: (1) post-shock recovery time (steps until population regains 90% of its pre-shock mean after standardized hub removal); (2) mean post-shock fragmentation; (3) per-capita costly-helping rate; (4) *story pull*: with $s_k(t)$ the value of macrostate component k at observer tick t , $b_k(t)$ the broadcast value, and $\varepsilon = 0.02$,

$$P = \left\langle \text{sign}(b_k(t) - s_k(t)) (s_k(t+1) - s_k(t)) \right\rangle_{(t,k) : |b_k(t) - s_k(t)| > \varepsilon}, \quad (1)$$

the average taken with equal weight over all qualifying (tick, component) pairs across the five broadcast components; $P > 0$ means the macrostate moves toward the broadcast where the two differ, and its magnitude is bounded by the largest per-tick component change. The threshold ε makes P well defined under false and noise broadcasts (and undefined for C, whose broadcast equals the current state). Campaigns: a *flagship* run (conditions A/B/C/F_{invert}/N $\times$ 50 paired seeds, mild shock), a *harsh-shock* replication (removal of 40% of hubs, $\times 30$ seeds), and a *sensitivity sweep* (feedback gain $g \in \{0.2, 0.8, 1.6\} \times \{A, C, F_{\text{invert}}, F_{\text{crisis}}, F_{\text{utopia}}, N\} \times 20$ seeds), a *scale check* re-running the core

Table 2. Primary paired comparisons (Wilcoxon signed-rank on per-seed differences; matched-pairs rank-biserial r ; bootstrap 95% CI of the median difference). FL = flagship (50 pairs), HS = harsh shock (30 pairs).

Outcome	Comparison	Δmedian [CI]	r	p
every outcome (FL, HS)	B vs A	all differences $\equiv 0$	—	—
story pull (FL)	F vs N	+0.0055 [0.0033, 0.0064]	0.84	1.1×10^{-8}
story pull (HS)	F vs N	+0.0065 [0.0043, 0.0078]	0.88	3.2×10^{-6}
story pull (HS)	R vs N	+0.0212 [0.0181, 0.0243]	1.00	1.9×10^{-9}
cooperation (FL)	C vs A	+0.200 [0.172, 0.239]	1.00	8.9×10^{-15}
cooperation (HS)	C vs A	+0.209 [0.170, 0.248]	0.99	9.3×10^{-9}
fragmentation (FL)	F vs A	-0.017 [-0.034, -0.006]	-0.53	9.5×10^{-4}
$\Delta\bar{\gamma}$ (HS)	F vs A	-0.336 [-0.456, -0.281]	-1.00	1.9×10^{-9}
$\Delta\bar{\gamma}$ (HS)	N vs A	-0.257 [-0.333, -0.218]	-0.98	1.9×10^{-8}
recovery time (FL, HS)	C, F, N vs A	$\Delta\text{median} = 0$ $ r < 0.5$		> 0.22

conditions with doubled population and resource capacity, a *response-regime campaign* repeating all broadcast conditions with conformist agents (15 seeds per cell), and a *replay control* (30 receiving runs paired with the harsh-shock seed list, plus 15 auxiliary source runs that generate the replayed trajectories): $250 + 150 + 360 + 90 + 32 + 30 + 15 = 927$ runs executed in total, each of 3,000–4,000 steps. Because conditions share seed lists, the design is paired; the primary analysis is a Wilcoxon signed-rank test on per-seed differences, reported with the matched-pairs rank-biserial correlation r and a bootstrap 95% CI on the median difference. Unpaired Mann–Whitney U with Cliff’s δ is retained as a robustness check and agrees with the paired analysis throughout; all other logged measures are exploratory.

5 Results

Table 2 summarizes the pre-specified comparisons; the following subsections unpack them.

5.1 Measurement alone is causally inert

Under paired seeds, condition B is trajectory-identical to A: every per-seed difference is exactly zero, on every outcome, in both the flagship and harsh-shock campaigns. The instrument acquires causal power only when it broadcasts — a validation that the design isolates the channel of interest.

5.2 A false self-model pulls the macrostate toward itself

Story pull (Eq. 1) under the inverted false broadcast exceeds the noise control at every gain tested (paired: flagship $\Delta\text{median} +0.0055$, $r = 0.84$, $p \approx 1.1 \times 10^{-8}$; harsh shock $+0.0065$, $r = 0.88$, $p \approx 3.2 \times 10^{-6}$). Decomposed by component,

the aggregate pull is carried chiefly by the turnover (≈ 0.033) and cooperation (≈ 0.005) components; the structural components are net-neutral, because the corrective over-connection documented below cancels drift toward the reported high fragmentation. Relative to noise the effect is strongest at *low* gain (ratio ≈ 7 at $g = 0.2$): a quiet, systematic lie out-steers loud noise, while at high gain noise itself begins to steer (Fig. 2, middle). Spillovers are substantial: the inverted broadcast reduces post-shock fragmentation relative to baseline (flagship paired $\Delta\text{median} -0.017$, $r = -0.53$, $p \approx 1 \times 10^{-3}$; directionally consistent but not significant in the harsh-shock replication, $p = 0.28$) because agents persistently told the network is fragmenting over-connect it (Fig. 3).

Plausibility grades the steering channel. A stronger structured control replaces the broadcast with a *replayed genuine self-model*: the recorded $S(t)$ trajectory of an observed-but-blind run with a different seed (condition R; 30 receiving runs sharing the harsh-shock seed list, each replaying one of 15 source trajectories) — realistic marginals and temporal coherence, no self-reference. The resulting steering hierarchy is $R > F > N$ (medians 0.024, 0.008, 0.002; paired over the 30 shared seeds, R vs N and F vs R both $|r| = 1.00$, $p \approx 1.9 \times 10^{-9}$). R is of course also a false description of the receiving network — merely a *plausible* one, since it was true of a sibling world. Steering power is therefore graded by plausibility: an achievable false self-model steers hardest, an adversarially inverted one still steers far more than unstructured noise, and content-free stimulation steers least. Signal structure matters, as one would expect — but the graded hierarchy shows the channel responds to *credible self-description*, not to persistence alone.

Structural consequences. Broadcasts densify and flatten the network. Relative to A (paired, harsh shock, post-transient medians), mean degree rises by +1.1 under truth ($r = 0.49$) and by +3.8/+4.2 under false/noise broadcasts ($r = 0.80/0.79$), while Freeman centralization falls (F vs A: -0.026 , $r = -0.74$) and betweenness concentration falls (F: -0.007 , $r = -0.67$; N similar): the fed-back network becomes denser and less hub-dominated than the silent one.

5.3 The type of lie sets the direction

The three distortion modes produce qualitatively different regimes (Fig. 2). The *inverted* self-model, always reporting the mirror of reality, generates the strongest pull: it keeps every corrective response permanently active and thereby parasitizes the population’s repair machinery. The *crisis* self-model is self-defeating for structure — corrections push fragmentation away from the alarmist report — yet its permanent cooperation-deficit signal mobilizes the highest costly-helping rates observed (0.36–0.45). The *utopia* self-model is behaviorally inert: because responses fire only on reported deficits, a flattering broadcast triggers nothing and is indistinguishable from disconnecting the channel, at every gain. Under purely corrective response rules, flattery equals silence.

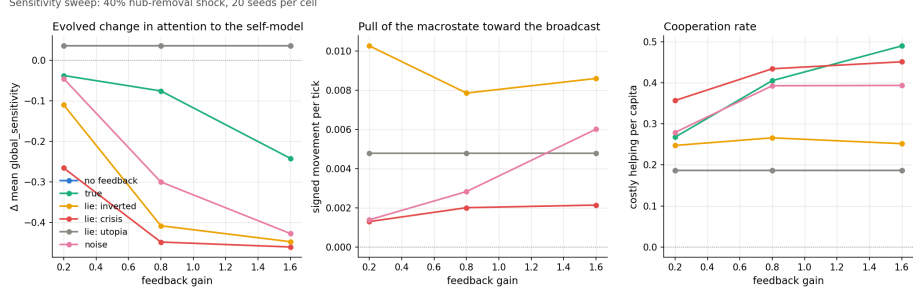


Fig. 2. Sensitivity sweep (40% hub-removal shock; 20 seeds per cell; medians). Left: evolved change in mean global sensitivity γ — attention to the self-model is selected against in proportion to broadcast unreliability and gain; the flattering (utopia) lie tracks the no-feedback baseline exactly. Middle: story pull toward the broadcast. Right: cooperation rate.

Table 3. The same lie under two response regimes (medians; harsh shock, $g = 0.8$). Baseline (A): cooperation 0.18, fragmentation 0.038.

Lie	Corrective		Conformist	
	coop.	fragm.	coop.	fragm.
Invert	0.27	0.021	0.42	0.054
Crisis	0.36	0.025	0.26	0.400
Utopia	0.19 (inert)	0.037 (inert)	0.428	0.025

5.4 The response regime decides which lies come true

Repeating all broadcast conditions with conformist agents reverses the taxonomy exactly as the mechanism predicts (Table 3). The *crisis* lie, self-defeating under correction, becomes strongly self-fulfilling under conformity: told the world is fragmenting, imitators cut ties and fragmentation rises an order of magnitude above baseline (0.400 vs 0.038). The *utopia* lie flips from behaviorally inert to benevolently self-fulfilling: “everyone cooperates” becomes true by imitation (cooperation 0.428, the highest observed in any condition). Under conformity noise steers as strongly as the inverted lie, since imitation amplifies any reported value. Which false self-models come true is therefore not a property of the lie alone but of the lie–response pair — the reflexive analogue of the observation that a measure’s distorting power depends on how actors respond to it.

5.5 Populations fed lies evolve to stop listening

Mean global sensitivity $\bar{\gamma}$ changes over ~ 40 generations as an emergent, unprogrammed function of broadcast quality (Fig. 4). At $g = 0.8$ (harsh-shock campaign): no broadcast -0.03 (neutral drift), truth -0.10 , noise -0.30 (paired $r = -0.98$), false -0.42 (paired Δ median -0.336 [$-0.456, -0.281$], $r = -1.00$,

The same world under two mirrors: identical seed, $t = 2000$

A — no feedback
mean degree 11.2 · cooperation 0.11 · fragmentation 0.000

F — false feedback (invert)
mean degree 22.4 · cooperation 0.21 · fragmentation 0.000

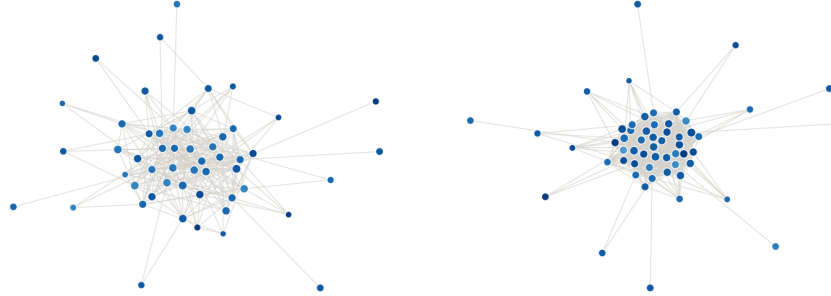


Fig. 3. The same world under two mirrors. Two runs from an identical seed at $t = 2000$: without feedback (left) and under the inverted false broadcast (right), which persistently reports fragmentation and thereby drives over-connection (mean degree 22.4 vs 11.2) and elevated costly helping. Node size encodes energy; color encodes generation.

$p \approx 1.9 \times 10^{-9}$). The sweep reveals a dose-response: distrust of the false broadcast deepens from -0.11 ($g = 0.2$) to -0.45 ($g = 1.6$) — and at the highest gain even *accurate* feedback is selected against (-0.24): a broadcast that perturbs behavior strongly becomes a fitness liability even when true. The utopia lie, being consequence-free, breeds no distrust at all.

5.6 Accurate feedback as a distributed homeostat

True feedback roughly doubles costly helping (flagship: 0.397 vs 0.179; paired Δ median +0.200 [0.172, 0.239], $r = 1.00$, $p \approx 9 \times 10^{-15}$; replicated under harsh shock, $r = 0.99$), rising monotonically with gain ($0.27 \rightarrow 0.49$); Fig. 5 shows the sustained divergence of trajectories. The loop functions as a population-level thermostat: reported deficits are repaired by distributed individual action.

5.7 A robust negative result

Post-shock recovery time was insensitive to feedback in every condition and campaign, even with 40% of hubs removed (all $p > 0.12$; median ≈ 40 steps). In this ecology, resilience is demographic — fast regrowth — rather than informational. We report this as a boundary of the phenomenon rather than adjusting the protocol until the null disappears.

5.8 Robustness to system scale

Re-running the core conditions (A, C, F_{invert} , N) with doubled initial population and proportionally scaled resources (steady-state population ≈ 120 –130 vs.

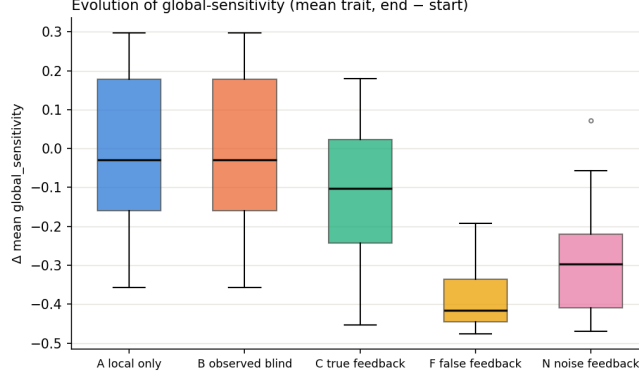


Fig. 4. Evolution of attention to the self-model (harsh-shock campaign, $g = 0.8$, 30 seeds per condition). Change in population-mean global sensitivity $\bar{\gamma}$ between the start and end of the run. The ordering (false < noise < true < none) emerged from selection alone.

≈ 60) preserves every effect under the paired analysis ($n = 8$ seed pairs per condition): cooperation under true feedback is 0.399 vs. 0.188 baseline (Δ median +0.151 [0.119, 0.247], $r = 1.00$, $p = 0.008$; 0.397/0.179 at reference scale), story pull retains F vs. N superiority (Δ median +0.0046 [0.0022, 0.0070], $r = 0.89$, $p = 0.023$), and the distrust ordering is intact with F vs. A at Δ median -0.299 [$-0.388, -0.212$], $r = -1.00$, $p = 0.008$. The phenomena are not finite-size artifacts of the reference configuration.

6 Discussion and Limitations

The first-order *directions* of response are encoded in the agents’ corrective rules; that is by design, and we do not claim them as discoveries. The findings live one level up: the magnitude and asymmetry of steering (steering graded by description plausibility; consistent lies $\gg$ matched noise, strongest at low gain), the spillovers to non-targeted structure, the qualitative taxonomy of lies (exploitative / mobilizing-but-self-defeating / inert), and above all the evolutionary layer, in which the credibility of a collective self-model becomes a selected property of the population. Both response regimes tested here are fixed rules; learned or heterogeneous response policies are the natural next step. Other limitations: a single ecology and parameter family (though effects are monotone across an $8 \times$ gain range and stable under a $2 \times$ scale change), rule-based rather than learned agents, and one network per run rather than interacting populations.

The results have a practical reading for engineered systems. Metric-driven infrastructure — autoscalers, observability pipelines, trending algorithms — constitutes exactly this loop; the false-broadcast conditions model telemetry corruption, and the utopia result implies that *flattering telemetry is operationally*

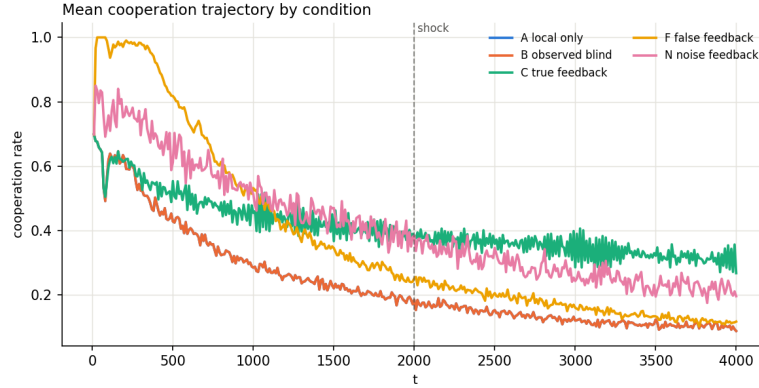


Fig. 5. Mean cooperation trajectories by condition (flagship campaign, 50 seeds per condition; dashed line marks the standardized shock). Conditions A and B coincide exactly. True and noise feedback sustain elevated cooperation; the inverted false broadcast sustains an intermediate level.

equivalent to no telemetry while raising no alarms. Documented episodes of measurement reshaping the measured system — from ranking reactivity in institutions [4] to the degradation of Google Flu Trends once its predictions entered the environment they forecast [18] — are natural empirical targets for the mechanisms isolated here, and connecting the model’s regimes to such datasets is the clearest path to empirical grounding. For multi-agent AI, where fleets of agents increasingly receive dashboards describing their own collective state, the evolution-of-distrust result suggests that persistent misreporting does not merely misdirect a population — it degrades the control channel itself.

7 Conclusion

We introduced a reproducible causal laboratory for reflexive self-modeling in adaptive networks and used it to show that (i) being measured changes nothing while being told changes much; (ii) a consistent false self-description steers a collective toward itself far more effectively than noise; (iii) the kind of lie determines whether it exploits, mobilizes, or pacifies; and (iv) selection acting on a heritable attention trait makes a population’s trust in its own self-model an evolving quantity — one that unreliable observers spend down. Future work extends the platform to continuity under total population turnover, learned response policies, and language-model agents.

Reproducibility. Code, configurations, all 912 stored run manifests plus the 15 replay-source trajectories, and one-command reproduction of every figure: github.com/Ivan825/Project-ONE.

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